

Sequential Aggregation-disaggregation Optimization Methods For Data Stream Mining

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Motivation

Algorithms for many optimization problems scale poorly for **large data**.

Standard Optimization Algorithm



Motivation

Algorithms for many optimization problems scale poorly for **large data**.

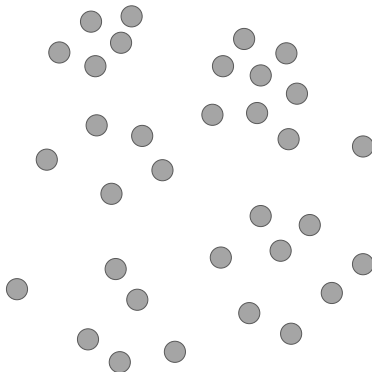
Standard Optimization Algorithm



Issues:

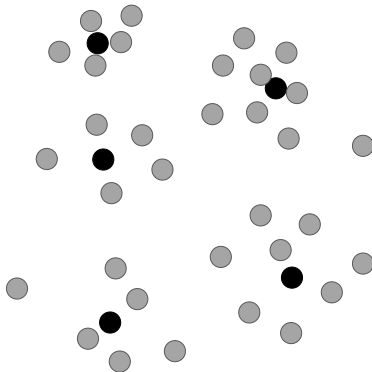
- Data does not fit into memory.
- Many iterations over the data.
- Typically have a super-linear run-time complexity.

Motivation



When data size is large, solving an optimization problem may be hard/intractable.

Motivation

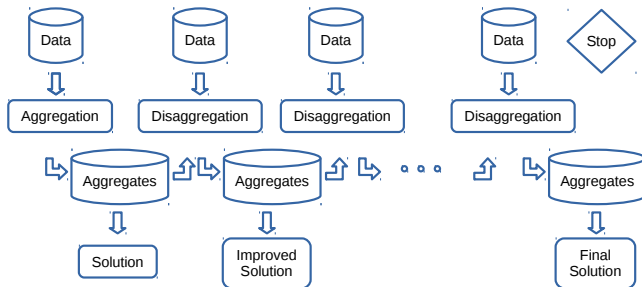


Can we optimize with aggregates?
Optimality?

Motivation

Iterative aggregation-disaggregation schemes have been shown to be effective for large data (Rogers et al, 1991; Park and Klabjan, 2016).

Iterative Aggregation/Disaggregation Framework



Iterative Aggregation-Disaggregation

The algorithms start by aggregating the original data, solving the problem on aggregated data, and then in subsequent steps gradually disaggregate the aggregated data to find a good (potentially optimal) solution.

Motivation

Data Stream

A data stream is a potentially unbounded sequence of observations. Processing stream is now common for many applications: GPS data from smart phones, web click-stream data, telecommunication connection data, readings from sensor nets, stock quotes.

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Data Stream

A data stream is a potentially unbounded sequence of observations. Processing stream is now common for many applications: GPS data from smart phones, web click-stream data, telecommunication connection data, readings from sensor nets, stock quotes.

Limited storage but potentially unbounded size of data streams pose the following challenges:

- ✓ Store only summaries (e.g., clusters).
- ✗ Real-time processing. Only a single pass over the data is possible.
- ✗ Concept drift: data distributions change over time.

Sequential Aggregation-Disaggregation

We propose a sequential aggregation-disaggregation optimization method where the disaggregation steps cannot be explicitly performed on past data. The method has the following properties:

- ① Anticipates disaggregation via partial aggregation.
- ② Performs partial aggregation sequentially as new data arrives.
- ③ Places more weight on newer data.

Motivation

Sequential Aggregation-Disaggregation

We propose a sequential aggregation-disaggregation optimization method where the disaggregation steps cannot be explicitly performed on past data. The method has the following properties:

- 1 Anticipates disaggregation via partial aggregation.
- 2 Performs partial aggregation sequentially as new data arrives.
- 3 Places more weight on newer data.

For data streams:

- ✓ Stores only summaries (e.g., clusters).
- ✓ Real-time processing. Only a single pass over the data.
- ✓ Follows changing distributions.

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IAD: Algorithm

Components of the algorithm: need to be tailored for a particular problem

- Definition aggregation/clustering procedure.
- Disaggregation procedure: How to partition the current clusters?
- Stopping/Optimality conditions

AID: algorithmic framework

Initialization: Define clusters and the aggregated data

While Stopping/Optimality condition is not satisfied

 Solve the problem with the aggregated data

 Check optimality condition / Decluster / Update the aggregated data

End While

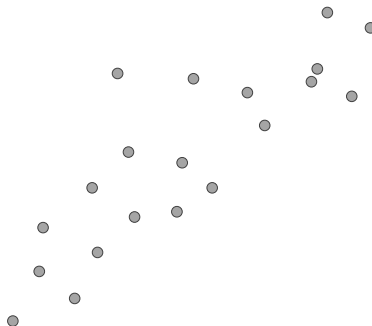
AID for LAD Regression

Least absolute deviation (LAD) regression

Given explanatory data $x \in \mathbb{R}^{n \times m}$ and response data $y \in \mathbb{R}^n$, find minimizer $\beta \in \mathbb{R}^m$

$$E^* = \min_{\beta \in \mathbb{R}^m} \sum_{i \in I} |y_i - \sum_{j \in J} x_{ij} \beta_j|$$

LAD illustration



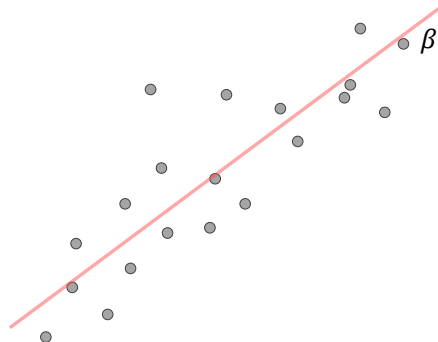
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LAD illustration



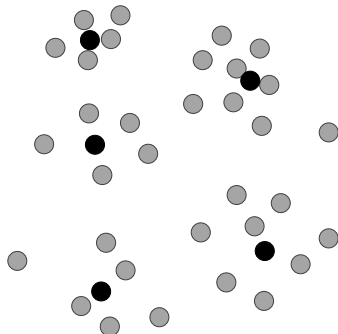
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Aggregated data: Average vector for each cluster



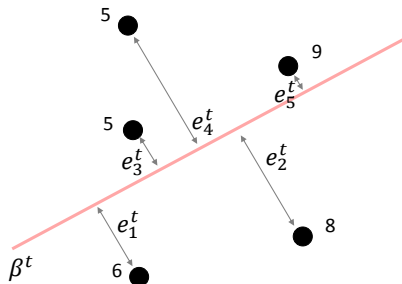
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Aggregated problem : Minimize $F^t = 6e_1^t + 8e_2^t + 5e_3^t + 5e_4^t + 9e_5^t$



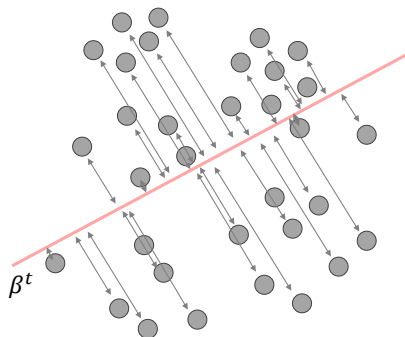
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Solution to the original problem: $E^t = \sum_{i=1}^n e_i$, where $e_i = |\beta^t x_i - y_i|$



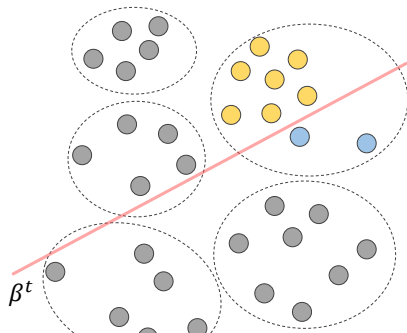
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$$E^* = \min_{\beta \in \mathbb{R}^m} \sum_{i \in I} |y_i - \sum_{j \in J} x_{ij} \beta_j|$$

Optimality condition: Are all observations in a cluster on the same side? (Park and Klabjan, 2016)



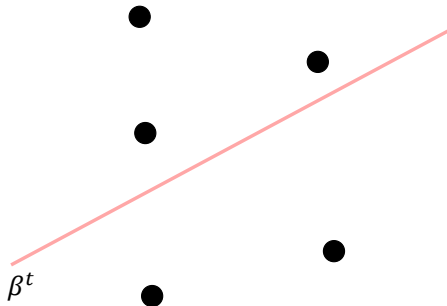
AID for LAD Regression: Illustration

While Optimality condition is not satisfied

Solve the problem with the aggregated data

Check optimality criteria and decluster

End While



Solve with the aggregated data

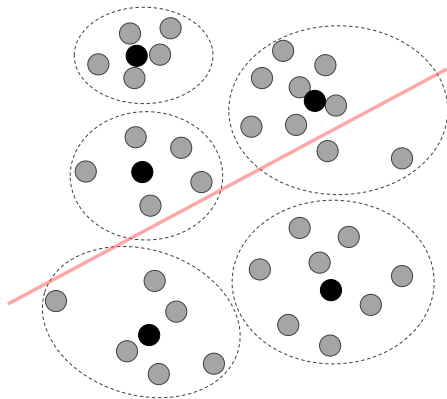
AID for LAD Regression: Illustration

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Check optimality criteria

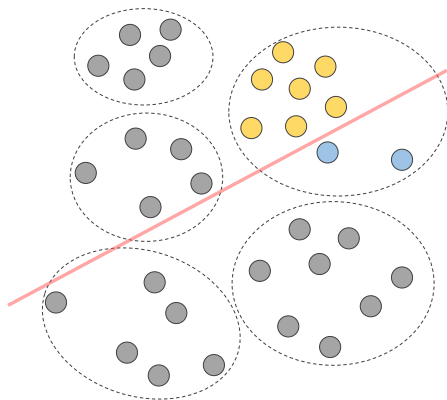
AID for LAD Regression: Illustration

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Decluster

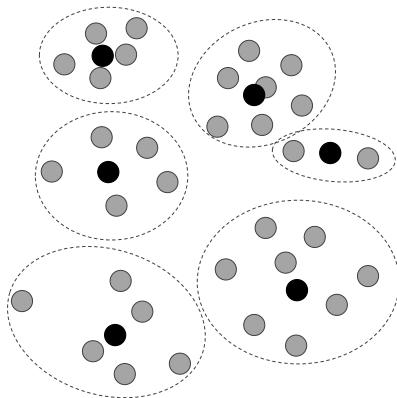
AID for LAD Regression: Illustration

While Optimality condition is not satisfied

Solve the problem with the aggregated data

Check optimality criteria and decluster

End While



Create new aggregated data

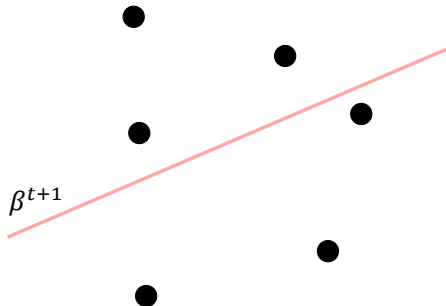
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While Optimality condition is not satisfied

Solve the problem with the aggregated data

Check optimality criteria and decluster

End While



Solve with the aggregated data

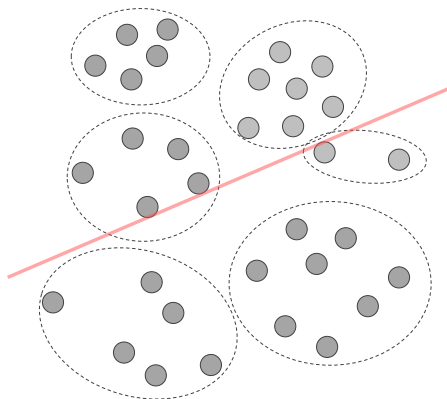
AID for LAD Regression: Illustration

While Optimality condition is not satisfied

Solve the problem with the aggregated data

Check optimality criteria and decluster

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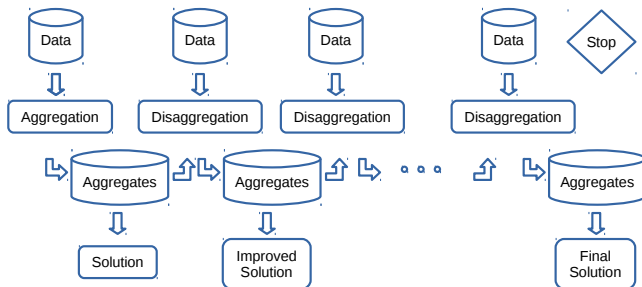
Check optimality criteria (optimal)

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Motivation

Iterative Aggregation/Disaggregation Framework

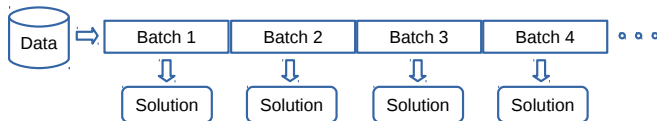


IAD is a powerful framework, but **needs repeated access to some data** to perform disaggregation. This is not possible for data streams.

Batch Processing

Why not just do batch processing?

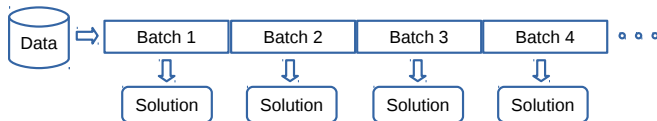
Batch Processing Framework



Batch Processing

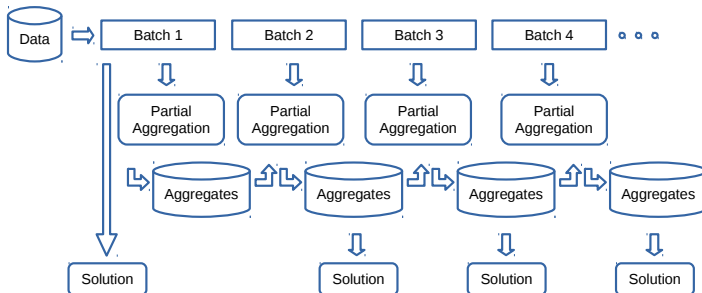
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Batch Processing Framework

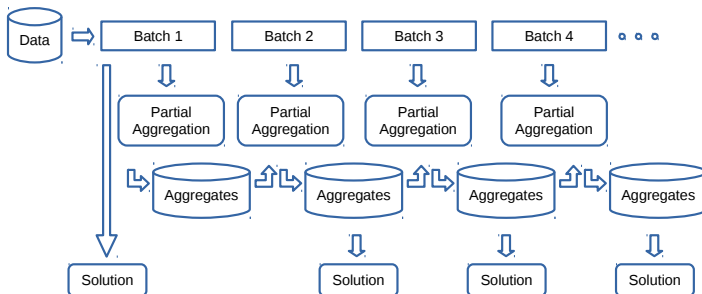


- Batch needs to be large enough to find a good solution.
- Information is not preserved over batches.
- Aggregating several solutions (e.g., by parameter averaging), does not optimize the overall objective function.

Partial Aggregation Framework for Streams



Partial Aggregation Framework for Streams



- **Partial aggregation:** Use a data stream clustering algorithm.

Example for LAD: Do not aggregate

- 1 points from different sides of the current regression line, and
- 2 points close to the current regression line.

Decay in data stream clustering will remove aggregation mistakes over time and allow the model to adapt to changes in the data.

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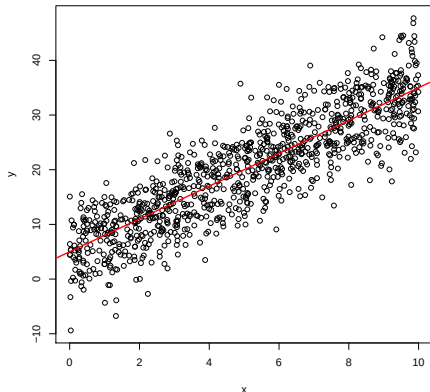
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Simple Data

1 million random data points for $x = [0, 10]$ following

$$y = 5 + 3x + \epsilon$$

with $\epsilon \sim N(0, 5)$.

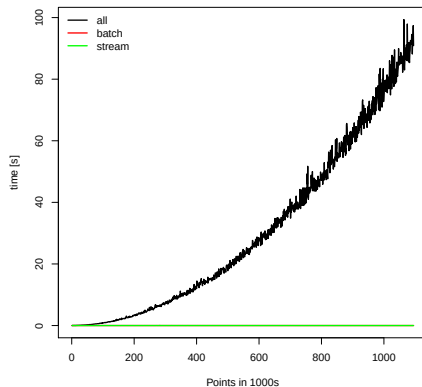
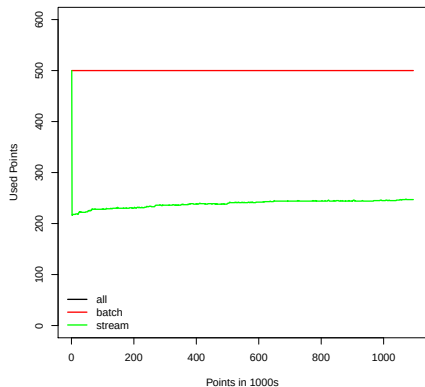


Simple Data Set

$n = 1$ million points

batch size $b = 500$

support points $s = 200$

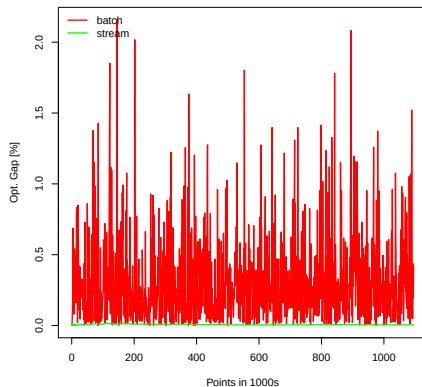
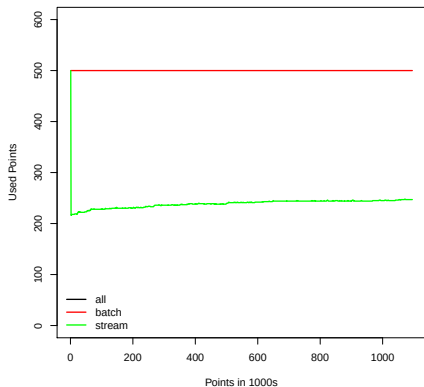


Simple Data Set

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batch size $b = 500$

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Difficult Data Set

- 1 million random data points, 10 dimensions
- True β_i , $i = \{1, 2, \dots, 10\}$, is randomly chosen from $\{-5, 5\}$.
- $x_i \sim N(\mu_i, \sigma_i)$ is a randomly generated feature where μ_i is uniformly chosen from $[-5, 5]$ and σ_i is chosen from $[1, 3]$.

$$y = \sum_{i=1}^{10} \beta_i x_i + \epsilon$$

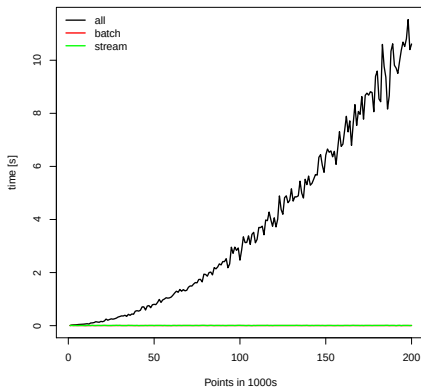
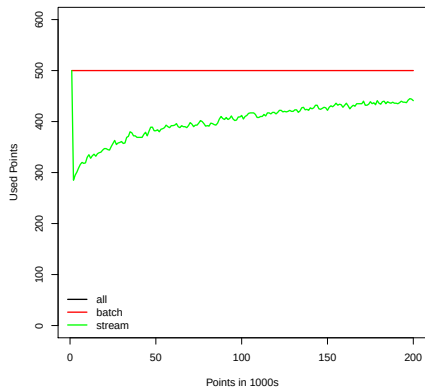
with $\epsilon \sim N(0, .2)$.

Difficult Data Set

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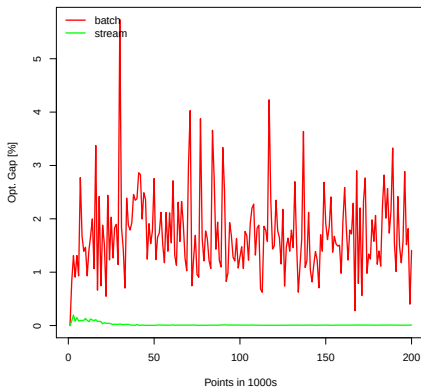
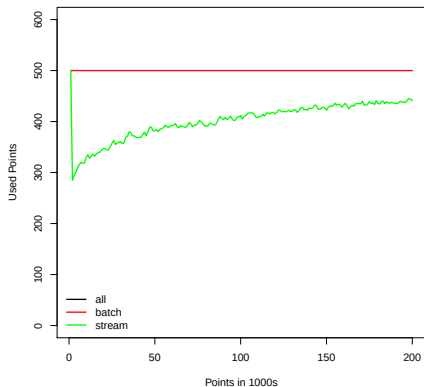


Difficult Data Set

$n = 1$ million points

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Conclusion and Future Work

Advantages:

- Partial aggregation anticipates future disaggregation needs.
- Partial aggregation is appropriate for data streams and leverages research from data stream clustering.
- Partial aggregation can help to improve quality over simple batch processing.

Future Work:

- Test different strategies to select which points should not be aggregated.
- Perform a comprehensive study.
- Apply the idea to other optimization problems (SVM, etc.).